SIRINDHORN INTERNATIONAL INSTITUTE OF TECHNOLOGY
SCHOOL OF INFORMATION, COMPUTER AND COMMUNICATION TECHNOLOGY
THAMMASAT UNIVERSITY

DES424 COURSE FINAL REPORT

InternFlow

Cloud-Based Centralized Internship Management & Tracking Platform

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Course: DES424 Cloud-based Application Development
Semester: Semester 1, Academic Year 2026
Presented to: Dr. Apichon Witayangkurn
Date: October 2026

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Degree Program: Digital Engineering (DE) / Computer Engineering (CPE)

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Date of Submission: October 2026

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Section 1: Revised Kick-Off Document

1.1 Project Overview & Background

Students and corporate employers coordinate internship recruitment and operational placements each semester. However, the existing process is predominantly managed through disconnected emails, spreadsheets, messaging applications (e.g., LINE, Discord), and paper forms. This ad-hoc approach creates substantial operational friction: applications are misplaced, candidate statuses remain ambiguous, working hours go unmonitored during the term, and performance assessments are often compiled hurriedly at the conclusion of the placement.

InternFlow is a modern, cloud-native, centralized internship and Cooperative and Work-Integrated Education (CWIE) management platform. It unifies the entire lifecycle of student internships into an accountable, transparent, and auditable digital environment.

Core Value Propositions

- **Centralized Corporate Job Board:** Verified employer postings featuring clear job descriptions, required competencies, department classifications, compensation/allowance figures, working hours, and work modality (`On-site` , `Remote` , `Hybrid`).
- **Multi-Stage Application & Interview Pipeline:** Real-time transparent tracking of candidate applications from initial submission, HR screening, interview scheduling (`INTERVIEW_SCHEDULED`), offer extension, single-placement exclusivity commitment, through company-approved cancellation.
- **Single Placement & Exclusivity Rule:** While candidates can receive multiple offers, they are limited to committing to exactly **ONE** official placement (`COMMITTED`), which automatically declines competing offers, eliminating candidate hoarding and placement abandonment.
- **Multi-Artifact Document Attachments:** Scalable file upload pipeline for CVs, portfolios, academic transcripts, and external project links (GitHub, Figma, Google Drive).
- **CWIE Operational Daily & Weekly Logbook:** Daily activity journaling with attendance modality tracking, pin-point task lists, problem-solving reflections, and automatic backdated submission detection.
- **Company Mentor Review & Sign-Off:** Supervisor inspection drawer enabling day-by-day and batch weekly approvals, 1–5 star performance ratings, and qualitative mentorship feedback.
- **Standardized 3-Dimensional Performance Rubric:** Final evaluation evaluating Work Quality, Punctuality & Responsibility, and Communication & Teamwork, yielding an official printable **Internship Completion Certificate & Performance Report**.
- **Seamless Multilingual Support:** Dynamic instant language switching between Thai (`TH`) and English (`EN`) across all system interfaces, forms, and status notifications.

1.2 Problem Statement & Current Shortcomings

The conventional internship recruitment and monitoring process suffers from several structural inefficiencies:

1. **Absence of a Single Source of Truth:** Students must scour disparate social media groups, university bulletin boards, and corporate career sites. There is no unified portal listing accredited positions with explicit allowance and modality terms.
2. **Untraceable Application Lifecycle:** Once an application is submitted via email or Google Forms, students have no visibility into its status. HR recruiters are burdened by repetitive inquiry emails, while students remain uncertain whether to accept competing opportunities.
3. **Absence of a Structured Interview Stage:** Informal communication regarding interview arrangements often leads to missed appointments, forgotten meeting links, and mismatched expectations between hiring managers and applicants.
4. **Offer Hoarding and Unannounced Dropouts:** Students frequently hold multiple offers simultaneously until the start date, subsequently dropping out without notice and leaving employers with unfilled positions.
5. **Unmonitored Daily/Weekly Progress:** Students typically compile paper logbooks or spreadsheets retroactively at the end of the internship period. Mentors are unable to verify attendance, daily deliverables, or technical roadblocks in real time.
6. **Superficial Performance Appraisals:** End-of-term evaluations are often completed as an administrative checkbox rather than a constructive, rubric-governed assessment.
7. **Fragmented Document Management:** Exchanging PDF resumes, portfolios, and signed agreement letters via email threads introduces security vulnerabilities, version confusion, and storage headaches.

1.3 Key Users and Stakeholders

The InternFlow platform is designed to support three primary user groups:

📐 ARCHITECTURE / WORKFLOW DIAGRAM (MERMAID)

```
graph LR
    classDef student fill:#e0f2fe,stroke:#0284c7,stroke-width:2px;
    classDef hr fill:#ecfdf5,stroke:#059669,stroke-width:2px;
    classDef admin fill:#fef3c7,stroke:#d97706,stroke-width:2px;

    User[InternFlow Platform]
    User ⟶ S[Student / Intern Candidate]:::student
    User ⟶ H[Company HR / Workplace Mentor]:::hr
    User ⟶ A[Institutional / University Administrator]:::admin

    S --- S1[Search Jobs & Apply]
    S --- S2[Attend Interviews & Commit to Offer]
    S --- S3[Log Daily CWIE Hours & Tasks]
    S --- S4[View Ratings & Print Report]

    H --- H1[Post & Manage Jobs]
    H --- H2[Screen & Schedule Interviews]
    H --- H3[Extend Offers & Manage Placements]
    H --- H4[Sign-Off Weekly Logs & Submit Rubric]

    A --- A1[Institutional Audit & CWIE Ledger]
    A --- A2[Monitor Student Placement Rates]
```

1. Students (Internship Seekers & Active Interns)

- Register and authenticate securely with email credentials.
- Maintain comprehensive academic profiles: University, Faculty, Major, Academic Year, GPA, Skills, Bio, and Avatar.
- Search and filter corporate internship postings by keyword, department, compensation, and attendance modality (`On-site` , `Remote` , `Hybrid`).
- Submit applications with personalized cover letters, multi-artifact attachments (PDF CVs, transcripts), and external URLs (GitHub, Figma).
- Track real-time application states: `PENDING` ➔ `REVIEWING` ➔ `INTERVIEW_SCHEDULED` ➔ `ACCEPTED` ➔ `COMMITTED` .
- Commit to exactly ONE official offer, triggering automatic withdrawal of all other pending/accepted submissions.
- Record daily log entries specifying working hours (default 8.0 hrs/day), modality, planned tasks, completed deliverables, troubleshooting notes, and learnings.
- Receive automatic warnings if a log entry is submitted retroactively (backdated detection).
- Monitor approved versus logged hours through live dashboard counters.
- Review supervisor sign-offs, 1–5 star ratings, and feedback.
- Generate and print official accredited Internship Completion Reports.

2. Company HR & Workplace Mentors (Corporate Employers)

- Maintain company profiles: Legal Name, Industry, Website, Office Address, Contact Email, Contact Phone, and Brand Logo.
- Create, publish, update, and toggle active status for internship job listings with compensation, schedules, and requirements.
- Review candidate rosters, inspect academic details, download attached documents, and view portfolio links.
- Advance candidate statuses through the recruitment pipeline: Reviewing ➔ Schedule Interview ➔ Extend Offer ➔ Reject.
- Review student placement cancellation requests with approval or decline workflows.
- Inspect active intern daily/weekly logs via an interactive slide-over drawer showing tasks, problems, solutions, and uploaded artifacts.
- Certify and sign off on log entries individually or batch-approve entire weekly periods with 1–5 star ratings and qualitative notes.
- Submit multi-dimensional performance rubrics (**Work Quality**, **Punctuality & Responsibility**, **Communication & Teamwork**) upon placement completion.

3. Institutional / University Administrators (Academic Coordinators)

- Oversee institutional CWIE student compliance and monitor placement percentages.
- Verify official internship certificates generated through the system with institutional audit integrity.

1.4 Functional Requirements (Comprehensive & Updated)

Subsystem 1: User Management & Authentication System

- **FR-1.1:** Secure registration and login using email and password credentials with server-side validation.
- **FR-1.2:** Passwords must be hashed using `bcrypt` (minimum 10 salt rounds) prior to storage; plaintext passwords must never be persisted.
- **FR-1.3:** Issue stateless JSON Web Tokens (JWT) signed with HMAC-SHA256 and a 24-hour expiration upon successful authentication.
- **FR-1.4:** Enforce Role-Based Access Control (RBAC) with predefined roles: `STUDENT` and `COMPANY_HR` .
- **FR-1.5:** Automatically initialize role-specific profile records (`StudentProfile` or `CompanyProfile`) upon user creation.
- **FR-1.6:** Mask sensitive academic fields (e.g., student GPA) from unauthorized public views, exposing them only to authenticated HR recruiters.
- **FR-1.7:** Enable public profile inspection of platform participants through slide-over modals (`PublicProfileModal`).

Subsystem 2: Job Listing & Discovery Management

- **FR-2.1:** Enable Company HR to create, edit, toggle active status, and delete internship postings.
- **FR-2.2:** Each posting must capture: Title, Department, Description, Requirements, Location, Work Modality (`isRemote`), Openings Count, Allowance/Compensation, Working Hours/Schedule, and Contact Information (Email, Phone, LINE ID).
- **FR-2.3:** Provide transactional cascade deletion of job postings and associated application records with explicit in-app confirmation modals.
- **FR-2.4:** Enable students to browse active job openings with full-text keyword search and multi-criteria filters (Department, Work Modality, Allowance).

Subsystem 3: Multi-Stage Application, Interview & Placement Pipeline

- **FR-3.1:** Enable students to apply for jobs with cover letters, direct file uploads (CVs, transcripts), and external URLs.
- **FR-3.2:** Strictly enforce the **One Active Application Per Company Rule** through database-level compound unique constraints (`studentId + jobPostId`).
- **FR-3.3 (Interview Stage):** Enable Company HR to transition candidates from `PENDING` or `REVIEWING` to `INTERVIEW_SCHEDULED` .
- **FR-3.4 (Interview Notice):** When an application enters `INTERVIEW_SCHEDULED` , the student's portal must display an Indigo Notice Banner instructing them that HR will coordinate date, time, and meeting details via email or phone.
- **FR-3.5 (Offer Extension):** Company HR can extend an official offer (`ACCEPTED`) either directly from review or following the interview stage.
- **FR-3.6 (Single Placement Commitment):** Students can commit to exactly ONE offer (`COMMITTED`). Committing automatically executes an atomic database transaction that sets all other active applications for that student to `REJECTED` / closed.
- **FR-3.7 (Controlled Cancellation Workflow):** A committed student cannot unilaterally abandon their placement. They must submit a cancellation request stating the reason (`CANCEL_REQUESTED`). The placement is only released once HR explicitly confirms (`CANCELLED`).
- **FR-3.8:** Complete Standardized Status Lifecycle: $$\text{PENDING} \longrightarrow \text{REVIEWING} \longrightarrow \text{INTERVIEW_SCHEDULED} \longrightarrow \text{ACCEPTED} \longrightarrow \text{COMMITTED} \underset{\text{Reject}}{\overset{\text{Approve}}{\rightleftarrows}} \text{CANCEL_REQUESTED} \longrightarrow \text{CANCELLED}$$

Subsystem 4: CWIE Daily & Weekly Operational Logbook

- **FR-4.1:** Access to the CWIE Logbook is strictly locked until a student possesses an active, confirmed placement (`COMMITTED` or `CANCEL_REQUESTED`).
- **FR-4.2:** Enable students to record daily logs specifying Date, Hours Worked (1.0–12.0 hours, default 8.0), Modality (`ON_SITE` , `REMOTE` , `HYBRID`), and Supervisor Name.
- **FR-4.3:** Capture itemized task lists (`tasksDone`), technical troubleshooting notes (`problemsAndSolutions`), key learnings, and attached artifacts (PDFs, code screenshots, project links).
- **FR-4.4 (Backdated Detection):** Automatically detect if a log entry date precedes the submission date; flag the entry as `isBackdated = true` and display a warning badge in both student and supervisor views.
- **FR-4.5:** Enforce uniqueness so a student can only create one log entry per calendar date (`@@unique([studentId, logDate])`).
- **FR-4.6:** Provide real-time calculation of total logged hours versus mentor-approved hours across the internship period.

Subsystem 5: Company Mentor Inspection & Sign-Off

- **FR-5.1:** Provide an interactive Inspector Drawer for Company HR/Mentors to examine intern daily logs, completed tasks, and artifact attachments.
- **FR-5.2:** Enable mentors to sign off on logs individually or execute batch one-click approvals for an entire week's logs.
- **FR-5.3:** Allow mentors to assign 1–5 star ratings and provide qualitative feedback notes that update the student's view in real time.

Subsystem 6: Performance Rubric Evaluation & Certification

- **FR-6.1:** Enable Company HR to submit an accredited 3-category performance rubric upon internship completion:
  1. **Work Quality & Technical Competence** (1–5 Stars)
  2. **Punctuality & Professional Responsibility** (1–5 Stars)
  3. **Communication & Team Collaboration** (1–5 Stars)
  4. **Qualitative Supervisor Remarks & Feedback**
- **FR-6.2:** Generate an official, print-ready (`window.print()`) **Internship Completion Certificate & Performance Report** consolidating student university metadata, verified hours, supervisor signatures, and rubric scores.

Subsystem 7: Bilingual Internationalization (TH / EN)

- **FR-7.1:** Full client-side internationalization with persistent language selection (`th` / `en`) via `LanguageContext` .
- **FR-7.2:** Instant UI translation across navigation menus, status badges, forms, modals, error toasts, and printable reports without page reloading.

1.5 Nonfunctional Requirements

1. Performance

- **NFR-1.1:** 95% of read API endpoints (`GET /api/jobs` , `GET /api/applications/my`) must respond within **< 500 ms** under normal load.
- **NFR-1.2:** Database queries must utilize indexed foreign keys and composite unique constraints to maintain query response times under **< 100 ms**.
- **NFR-1.3:** File uploads (up to 10 MB) must complete streaming to Amazon S3 within **< 5 seconds** over standard broadband connections.

2. Scalability

- **NFR-2.1:** The backend must be stateless to permit horizontal scaling via Amazon ECS Fargate container auto-scaling based on CPU/memory utilization thresholds (70%).
- **NFR-2.2:** Database connection pooling managed via Prisma Client to sustain at least **200 concurrent active sessions** without connection exhaustion.
- **NFR-2.3:** Frontend static bundle served via Amazon CloudFront edge locations with global caching to handle sudden traffic spikes effortlessly.

3. Reliability & Availability

- **NFR-3.1:** Target service availability of **99.5% uptime** during academic terms.
- **NFR-3.2:** Multi-table writes (such as offer acceptance and competing application decline) must execute within atomic database transactions (`prisma.$transaction`) to prevent orphaned records.
- **NFR-3.3:** Amazon RDS Multi-AZ deployment ensures automated failover to a standby replica in a secondary Availability Zone in **< 60 seconds** in case of infrastructure failure.

4. Data Integrity & Consistency

- **NFR-4.1:** Database foreign key constraints enforce referential integrity across users, profiles, jobs, applications, logs, and evaluations.
- **NFR-4.2:** Server-side validation rejects invalid status transitions (e.g., transitioning from `REJECTED` directly to `COMMITTED`).

5. Security & Privacy

- **NFR-5.1:** Passwords must be hashed using `bcrypt` (10 rounds). Plaintext passwords must never be logged or persisted.
- **NFR-5.2:** All protected routes require a valid JWT passed in the HTTP `Authorization: Bearer <token>` header.
- **NFR-5.3:** Sensitive student metadata (GPA) is strictly masked from unauthorized users.
- **NFR-5.4:** Amazon S3 buckets are configured with Block Public Access enabled; files are accessed via authenticated backend proxies or pre-signed URLs.
- **NFR-5.5:** Cross-Origin Resource Sharing (CORS) is configured to restrict API access exclusively to authorized frontend domains.

6. Usability & Accessibility

- **NFR-6.1:** Modern, responsive UI adhering to the Emerald & Mint SaaS design system, fully functional on desktop, tablet, and mobile displays (minimum viewport width 360px).
- **NFR-6.2:** Immediate visual feedback provided for all user actions via skeleton loaders, spinners, and interactive toast notifications.

1.6 Core Programming Languages

- **TypeScript (v5.x):** Utilized across both Frontend and Backend codebases to enforce end-to-end type safety, eliminate runtime type errors, and improve code maintainability.
- **JavaScript (ES2023+):** Underlying language for Node.js runtime execution.
- **HTML5:** Semantic markup foundation for web standards, responsive layouts, and print-media stylesheets.
- **CSS3 / Modern CSS:** Component styling powered by Tailwind CSS v4 utility classes and CSS variables.
- **SQL (PostgreSQL Dialect):** Relational query language executed via Prisma 7 ORM.

1.7 Cloud Infrastructure

Main Components (Standalone Managed Cloud Services)

Component Name	Service Category	Role in InternFlow Architecture
Amazon Route 53	Networking & DNS	Highly available Cloud Domain Name System (DNS) providing domain resolution, latency routing, and health checks.
Amazon CloudFront	Content Delivery Network	Global CDN caching frontend static assets (HTML, JS, CSS, images) at edge locations for low-latency worldwide delivery.
Amazon Simple Storage Service (S3)	Object Storage	Scalable bucket storage for hosting static frontend builds and securely storing student CVs, transcripts, avatars, and logbook artifacts.
Amazon Elastic Container Service (ECS)	Container Orchestration	Serverless container management via **AWS Fargate**, running backend Node.js/Express containers with auto-scaling and zero server administration.
Amazon Elastic Container Registry (ECR)	Container Registry	Secure Docker container image repository for storing versioned production backend images.
Amazon Relational Database Service (RDS)	Managed Database	Fully managed PostgreSQL 16 database configured with Multi-AZ automated replication, automated backups, and storage encryption.
AWS Identity and Access Management (IAM)	Security & Identity	Enforces granular least-privilege security policies, execution roles for ECS tasks, and secure S3 access credentials.
Amazon CloudWatch	Monitoring & Observability	Centralized log ingestion from ECS containers, metric alarms for CPU/memory utilization, and API error tracking.

Micro Components (Infrastructure & Networking Primitives)

Component Name	Service Category	Role in InternFlow Architecture
Amazon Virtual Private Cloud (VPC)	Isolated Network	Dedicated virtual network logically isolating backend compute clusters and database instances from the public internet.
Public & Private Subnets	Subnet Segmentation	Multi-AZ architecture: Public subnets host Load Balancers; Private subnets host ECS container tasks; Isolated DB subnets host RDS instances.
Application Load Balancer (ALB)	Traffic Distribution	Distributes incoming HTTPS traffic across healthy ECS Fargate container tasks with automated health check polling on `/api/health`.
Security Groups	Virtual Firewalls	Stateful packet filtering controlling inbound and outbound traffic at the port level (ALB allows port 443; ECS allows port 4000 from ALB only; RDS allows port 5432 from ECS only).
Amazon S3 VPC Endpoint (Gateway)	Private Networking	Enables direct, private routing of S3 file upload traffic from ECS containers without traversing the public internet or incurring NAT gateway charges.

1.8 Tech Stack & Tooling Matrix

Layer / Concern	Technology Selected	Version / Spec	Justification & Role
Frontend Framework	React	19.x with Vite 6	High-performance SPA with instant HMR and optimized component rendering
Frontend Routing	React Router	7.x	Declarative client-side routing with nested layouts and route protection
Styling Engine	Tailwind CSS	v4.0	Modern utility-first CSS framework with tailored color tokens and responsive utilities
Icons & Visuals	Lucide React	Latest	Clean, consistent SVG icon set matching modern SaaS aesthetics
Backend Runtime	Node.js	v20+ LTS	Event-driven, asynchronous I/O optimized for concurrent web APIs
Web Framework	Express.js	5.x	Minimalist, robust REST API routing with native async error handling
Database Engine	PostgreSQL	16.x (Multi-AZ)	Enterprise-grade ACID relational database with strong JSON support
ORM / Data Access	Prisma	7.x	Type-safe query builder, automated migrations, and schema modeling
Authentication	JWT + bcryptjs	24h Expiry	Stateless token-based auth with secure password hashing (10 salt rounds)
File Upload Pipeline	Multer + multer-s3	AWS SDK v3	Multipart form data streaming directly to S3 with local disk fallback
Internationalization	React Context API	`LanguageContext`	Zero-dependency, lightweight real-time bilingual switching (TH / EN)
Containerization	Docker & Compose	Multi-stage build	Reproducible environments across local development and production
Testing Tools	Playwright & Robot Framework	Latest	Automated End-to-End browser test automation and API validation
Version Control	Git & GitHub	Trunk / Feature branch	Collaborative source control, pull request reviews, and issue tracking

---

Section 2: Software Architecture (Lecture 3 Topics)

2.1 Subsystem Decomposition & Layered Architectural Style

In accordance with software engineering principles (Lecture 3: System Decomposition into Cohesive Subsystems with Defined Interfaces), the InternFlow platform is structured into **10 cohesive subsystems** organized across **5 logical architectural layers**.

📐 ARCHITECTURE / WORKFLOW DIAGRAM (MERMAID)

```
graph TD
    classDef pres fill:#eff6ff,stroke:#3b82f6,stroke-width:2px;
    classDef api fill:#f0fdf4,stroke:#22c55e,stroke-width:2px;
    classDef domain fill:#fdf4ff,stroke:#d946ef,stroke-width:2px;
    classDef data fill:#fffbeb,stroke:#f59e0b,stroke-width:2px;
    classDef infra fill:#f8fafc,stroke:#64748b,stroke-width:2px;

    subgraph Layer 1: Presentation Layer
        UI_Nav[Navigation & Responsive Layout]:::pres
        UI_Dash[Role-Based Dashboards]:::pres
        UI_Jobs[Job Search & Board UI]:::pres
        UI_App[Application Pipeline & Stepper UI]:::pres
        UI_Log[CWIE Daily Logbook & Drawer UI]:::pres
        UI_Cert[Printable Certificate UI]:::pres
        UI_Lang[Bilingual Language Context]:::pres
    end

    subgraph Layer 2: API Gateway & Controller Layer
        API_Auth[Auth Controller: /api/auth]:::api
        API_Users[User Profile Controller: /api/users]:::api
        API_Jobs[Job Controller: /api/jobs]:::api
        API_Apps[Application Controller: /api/applications]:::api
        API_Log[Logbook Controller: /api/logbook]:::api
        API_Eval[Evaluation Controller: /api/evaluations]:::api
    end

    subgraph Layer 3: Domain & Business Logic Layer
        DOM_Auth[Authentication & RBAC Enforcement Service]:::domain
        DOM_Commit[Single-Placement Exclusivity & Auto-Decline Engine]:::domain
        DOM_Log[CWIE Backdated Detection & Hour Aggregation Engine]:::domain
        DOM_Eval[Rubric Scoring & Certification Engine]:::domain
        DOM_File[S3 File Sanitization & Upload Service]:::domain
    end

    subgraph Layer 4: Data Access & Persistence Layer
        Prisma_Client[Prisma 7 ORM Query Engine]:::data
        DB_Tables[(PostgreSQL 16 Relational Tables)]:::data
        S3_Storage[(Amazon S3 Object Storage)]:::data
    end

    subgraph Layer 5: Cloud Infrastructure Layer
        AWS_Route53[Route 53 DNS]:::infra
        AWS_CF[CloudFront CDN]:::infra
        AWS_ALB[Application Load Balancer]:::infra
        AWS_ECS[ECS Fargate Clusters]:::infra
        AWS_RDS[RDS Multi-AZ PostgreSQL]:::infra
    end

    UI_Nav ⟶ API_Auth
    UI_Jobs ⟶ API_Jobs
    UI_App ⟶ API_Apps
    UI_Log ⟶ API_Log
    UI_Cert ⟶ API_Eval

    API_Auth ⟶ DOM_Auth
    API_Apps ⟶ DOM_Commit
    API_Log ⟶ DOM_Log
    API_Eval ⟶ DOM_Eval

    DOM_Auth ⟶ Prisma_Client
    DOM_Commit ⟶ Prisma_Client
    DOM_Log ⟶ Prisma_Client
    DOM_Eval ⟶ Prisma_Client
    DOM_File ⟶ S3_Storage

    Prisma_Client ⟶ DB_Tables
```

Subsystem Breakdown:

1. **User Authentication & Identity Subsystem:** Handles account registration, bcrypt password hashing, credential verification, stateless JWT issuance, token decoding, and Role-Based Access Control (RBAC) route guarding.
2. **Profile & Academic Metadata Subsystem:** Manages student academic information (GPA, faculty, major, skills, avatar) and company corporate details. Implements academic privacy rules (masking student GPA from public view).
3. **Job Listing & Discovery Subsystem:** Provides CRUD capabilities for corporate internship listings, multi-criteria filtering (department, modality, allowance), and public job viewing.
4. **Application & Placement Pipeline Subsystem:** Manages candidate application submissions, enforces the one-active-application-per-company rule, coordinates the interview stage (`INTERVIEW_SCHEDULED`), and executes the atomic single-placement commitment rule (`COMMITTED`) with automatic competing offer withdrawal.
5. **CWIE Daily Logbook Subsystem:** Manages daily activity records, modality tracking (`ON_SITE` , `REMOTE` , `HYBRID`), itemized tasks, and automated backdated submission auditing.
6. **Company Mentor Review & Sign-Off Subsystem:** Renders the supervisor inspector drawer, enabling mentors to examine deliverables, award 1–5 star ratings, and record qualitative feedback for individual days or entire weeks.

7. **Performance Rubric Evaluation Subsystem:** Governs end-of-term evaluations across Work Quality, Punctuality, and Teamwork; calculates aggregated scores; and formats printable completion certificates.
8. **Media & Artifact Storage Subsystem:** Manages multi-part file uploads (PDF CVs, transcripts, avatar images) via Multer and AWS SDK v3 streaming directly to Amazon S3 with collision-resistant timestamp key prepending.
9. **Bilingual Localization Subsystem:** Manages client-side translation dictionaries (`th` / `en`) via React Context to render UI components instantly in the selected language.
10. **System Observability & Monitoring Subsystem:** Implements health probes (`GET /api/health`), structured JSON error formatting, and streaming application metrics and logs to Amazon CloudWatch.

2.2 Data Management (Lecture 3 Topics)

Lecture 3 emphasizes structured data management: distinguishing persistent data, transient data, storage media, and concurrency control.

📐 ARCHITECTURE / WORKFLOW DIAGRAM (MERMAID)

```
graph TD
    subgraph Client Memory
        C1[Auth Token in localStorage]
        C2[Active Form State / Filter State]
        C3[Current Language Selection]
    end

    subgraph Backend Memory
        B1[Decoded JWT Payload in Request Context]
        B2[Multer Memory Stream Buffers]
        B3[Prisma Connection Pool Socket Cache]
    end

    subgraph Persistent Storage: Amazon RDS PostgreSQL 16
        D1[(User & Authentication Records)]
        D2[(Student & Company Profiles)]
        D3[(Job Posts & Application States)]
        D4[(CWIE Daily Logs & Mentor Sign-Offs)]
        D5[(Rubric Evaluations & Scores)]
    end

    subgraph Persistent Storage: Amazon S3 Object Storage
        S1[(Student Resumes & CVs)]
        S2[(Academic Transcripts)]
        S3[(Company Logos & Avatars)]
        S4[(Logbook Deliverable Artifacts)]
    end

    Client Memory -. HTTP Requests .-> Backend Memory
    Backend Memory ⟶ Persistent Storage: Amazon RDS PostgreSQL 16
    Backend Memory ⟶ Persistent Storage: Amazon S3 Object Storage
```

1. Persistent Data Management

- **Database Engine:** Amazon RDS PostgreSQL 16 Multi-AZ.
- **Relational Tables:** Persists structured business entities across 8 relational tables (`User` , `StudentProfile` , `CompanyProfile` , `JobPost` , `Application` , `Document` , `DailyLog` , `WeeklyLog` , `InternshipEvaluation`).
- **Object / Blob Storage:** Binary documents (PDF resumes, transcripts, avatar images, code screenshots) are stored in private Amazon S3 buckets with path prefixes (`/resumes/` , `/transcripts/` , `/avatars/`). The database stores only the immutable S3 URLs (`fileUrl`).
- **Schema Evolution:** Managed through declarative, version-controlled Prisma migrations (`prisma migrate dev` / `prisma migrate deploy`), ensuring reproducible database structures across development, staging, and production environments.

2. Transient Data Management

- **Client Session State:** Authentication tokens (JWT) and user identity profiles are stored in browser `localStorage` .
- **Request Scope Data:** In-flight HTTP requests carry decoded user identity (`req.user = { userId, role }`) populated by the `authenticate` middleware in memory.
- **File Upload Buffers:** File uploads are streamed through memory-efficient streaming buffers directly to S3 via `multer-s3` , avoiding local disk consumption in production container environments.

3. Concurrency Control & Transactional Integrity

- **Atomic Offer Acceptance:** When a student accepts an offer, the system must update the chosen application to `COMMITTED` while updating all other active applications for that student to `REJECTED` . This is wrapped inside an atomic `prisma.$transaction` :

```
await prisma.$transaction(async (tx) => {
  // 1. Commit target placement
  await tx.application.update({
    where: { id: targetAppId },
    data: { status: 'COMMITTED' }
  });
  // 2. Automatically decline competing applications
  await tx.application.updateMany({
    where: {
      studentId: student.id,
      id: { not: targetAppId },
      status: { in: ['PENDING', 'REVIEWING', 'INTERVIEW_SCHEDULED', 'ACCEPTED'] }
    },
    data: { status: 'REJECTED' }
  });
});
```

- **Database Constraints:**
  - Candidate spamming prevented by `@@unique([studentId, jobPostId])`.
  - Duplicate daily logging prevented by `@@unique([studentId, logDate])`.
  - Foreign key cascade deletion cleans up child records when a parent entity is deleted.

4. Backup & Disaster Recovery

- **Automated Daily Snapshots:** Amazon RDS executes automated daily backups with a 35-day retention window.
- **Point-in-Time Recovery (PITR):** Transaction logs (WAL) are streamed continuously, enabling restoration to any specific second within the retention window.
- **S3 Bucket Versioning:** S3 buckets maintain object versioning to protect against accidental deletion or overwrite.

2.3 Boundary Conditions (Lecture 3 Topics)

Boundary conditions define system behavior during non-steady states: system initialization, termination/shutdown, and unexpected failure modes.

1. System Initialization

- **Cloud Infrastructure Bootstrapping:**
  1. Route 53 routes incoming DNS queries to CloudFront (frontend) and ALB (backend).
  2. The Application Load Balancer begins polling the health probe endpoint (`GET /api/health`).
  3. ECS Fargate pulls container images from Amazon ECR and launches tasks within private subnets.
  4. Node.js backend initializes: loads environment variables (`.env`), connects to PostgreSQL via Prisma connection pool, verifies schema migrations, and binds to port 4000.
  5. The ALB marks targets as healthy once HTTP 200 is confirmed, opening traffic routing.

- **Client Application Bootstrapping:**
  1. The browser requests `index.html` via CloudFront CDN.
  2. React hydrates and initializes `LanguageContext` (restoring stored language from `localStorage` or defaulting to `th`).
  3. The auth provider inspects `localStorage` for an existing JWT token. If present and valid, it loads user state; if expired or invalid, it purges stored credentials and directs the user to `/login`.

2. System Termination

- **Graceful Container Termination (ECS Scale-In / Deployments):**
  1. AWS ECS sends a `SIGTERM` signal to the Node.js process.
  2. The Express server stops accepting new incoming connections on port 4000.
  3. ALB deregisters the target container, routing new traffic to other active tasks.
  4. Active in-flight HTTP requests are given a 30-second draining window to complete.
  5. The Prisma connection pool is gracefully disconnected (`await prisma.$disconnect()`).
  6. The Node.js process exits cleanly (`exit code 0`).

- **User Session Termination (Logout):**
  1. User clicks "Sign Out" (`ออกจากระบบ`).
  2. Frontend immediately removes `token` and `user` keys from browser `localStorage`.
  3. Client authentication state is reset to `null`.
  4. React Router navigates the user to `/login` with an in-app confirmation toast.

3. Failure Modes & Automated Recovery

- **Primary Database Outage:**
  - *Symptom:* Primary RDS PostgreSQL instance becomes unresponsive.
  - *Automated Recovery:* Amazon RDS Multi-AZ automatically detects loss of heartbeat and initiates automatic failover to the synchronous standby replica in the secondary Availability Zone. The database DNS endpoint is updated within < 60 seconds without data loss. Prisma connection pools detect dropped connections and reconnect with exponential backoff.

- **Backend Container Crash:**
  - *Symptom:* Uncaught exception or OOM event crashes a Node.js process.
  - *Automated Recovery:* ECS Fargate task supervisor detects process termination and automatically spawns a replacement container task. The ALB diverts traffic away from the failing task to surviving instances, maintaining zero service downtime.

- **API Error Handling & Stack Trace Masking:**

- *Behavior:* Centralized Express error-handling middleware catches all unhandled rejections, logs full error details and stack traces to Amazon CloudWatch, and returns sanitized, user-friendly JSON responses (`{ error: "Internal server error" }` , HTTP 500) to protect internal infrastructure details.

- **Network Disconnection in Client Browser:**
  - *Behavior:* Frontend API client detects network timeouts and surfaces an alert toast ("Unable to connect to server. Please try again.") while preserving form state in component memory, preventing user data loss.

2.4 Access Control & Security Architecture

Role-Based Access Control (RBAC) Matrix

Resource / Action	Endpoint	Unregistered Guest	Student (Applicant)	Student (Placed Intern)	Company HR / Mentor
Register / Sign In	`POST /api/auth/*`	**Allow**	**Allow**	**Allow**	**Allow**
Browse / Search Jobs	`GET /api/jobs`	**Allow**	**Allow**	**Allow**	**Allow**
Create / Edit / Delete Job	`POST, PUT, DELETE /api/jobs/*`	Deny	Deny	Deny	**Allow**
Submit Job Application	`POST /api/applications`	Deny	**Allow**	Deny *(Committed Lock)*	Deny
View Submitted Applications	`GET /api/applications/my`	Deny	**Allow**	**Allow**	Deny
Schedule Interview	`PATCH /api/applications/:id/status`	Deny	Deny	Deny	**Allow** (`INTERVIEW_SCHEDULED`)
Extend / Reject Offer	`PATCH /api/applications/:id/status`	Deny	Deny	Deny	**Allow** (`ACCEPTED` , `REJECTED`)
Accept Offer & Commit	`PATCH /api/applications/:id/commit`	Deny	**Allow** *(Single placement)*	Deny *(Already committed)*	Deny
Request Placement Cancellation	`POST /api/applications/:id/cancel`	Deny	Deny	**Allow** (`CANCEL_REQUESTED`)	Deny
Approve / Decline Cancellation	`PATCH /api/applications/:id/cancellation-action`	Deny	Deny	Deny	**Allow**
Create / Edit CWIE Daily Log	`POST, PUT /api/logbook/*`	Deny	Deny *(Locked)*	**Allow**	Deny
Sign-Off Weekly Logs (1-5★)	`PATCH /api/logbook/signoff/*`	Deny	Deny	Deny	**Allow**
Submit Final Evaluation Rubric	`POST /api/evaluations`	Deny	Deny	Deny	**Allow**
View / Print Completion Certificate	`GET /api/evaluations/student/:id`	Deny	Deny	**Allow**	**Allow**
View Masked GPA	`GET /api/users/profile/:id`	Deny *(Masked)*	**Allow** *(Self only)*	**Allow** *(Self only)*	**Allow** *(Recruiter view)*

📐 ARCHITECTURE / WORKFLOW DIAGRAM (MERMAID)

```
graph TD
    classDef client fill:#e0f2fe,stroke:#0284c7,stroke-width:2px;
    classDef mw fill:#fef3c7,stroke:#d97706,stroke-width:2px;
    classDef route fill:#ecfdf5,stroke:#059669,stroke-width:2px;
    classDef err fill:#fee2e2,stroke:#ef4444,stroke-width:2px;

    Req[Incoming HTTP Request]:::client
    Req --> H{Has Authorization: Bearer token?}

    H -- No --> CheckPublic{Is Public Route?}
    CheckPublic -- Yes --> ExecPublic[Execute Public Handler]:::route
    CheckPublic -- No --> Ret401[Return 401 Unauthorized]:::err

    H -- Yes --> VerifyJWT{Verify JWT Signature & Expiry}:::mw
    VerifyJWT -- Invalid / Expired --> Ret401
    VerifyJWT -- Valid --> ExtractUser[Populate req.user = payload]:::mw

    ExtractUser --> CheckRole{Does req.user.role match required roles?}:::mw
    CheckRole -- No --> Ret403[Return 403 Forbidden]:::err
    CheckRole -- Yes --> ExecHandler[Execute Controller & Service Logic]:::route
```

Security Measures Implemented:

1. **Password Encryption:** Passwords salted and hashed with `bcryptjs` using 10 rounds.
2. **Stateless JWT Authorization:** Tokens carry cryptographically signed user IDs and roles with a 24-hour expiration window.
3. **Parameter Tampering Prevention:** Status modifications on applications verify that the calling HR belongs to the company that posted the job, and candidate commits verify that the caller is the student who applied.
4. **SQL Injection Defense:** All database interactions are executed via Prisma ORM parameterized queries.
5. **Cross-Site Scripting (XSS) Prevention:** React auto-escapes rendered text; cover letter and task notes are stored and rendered as sanitized strings.

Section 3: Detailed Design

3.1 Entity-Relationship (ER) Diagram & Schema Specification

The complete relational schema for InternFlow is modeled in PostgreSQL 16 via Prisma 7:

📐 ARCHITECTURE / WORKFLOW DIAGRAM (MERMAID)

```
erDiagram
    User ||--o| StudentProfile : "has profile"
    User ||--o| CompanyProfile : "has profile"
    User ||--o{ Document : "owns"

    StudentProfile ||--o{ Application : "submits"
    StudentProfile ||--o{ DailyLog : "records"
    StudentProfile ||--o{ WeeklyLog : "records"
    StudentProfile ||--o{ InternshipEvaluation : "evaluated in"

    CompanyProfile ||--o{ JobPost : "publishes"
    CompanyProfile ||--o{ InternshipEvaluation : "evaluates"

    JobPost ||--o{ Application : "receives"

    Application ||--o{ Document : "attaches"
    Application ||--o{ DailyLog : "accumulates"
    Application ||--o{ WeeklyLog : "accumulates"
    Application ||--o| InternshipEvaluation : "results in"

    User {
        String id PK "UUID"
        String email UK "Unique Email"
        String password "bcrypt hash"
        String name "Full Name"
        Role role "STUDENT | COMPANY_HR | UNIVERSITY_ADMIN"
        String avatarUrl "S3 URL"
        DateTime createdAt
        DateTime updatedAt
    }

    StudentProfile {
        String id PK "UUID"
        String userId FK "Unique 1:1"
        String studentId "Student ID Number"
        String university "Institution Name"
        String faculty "Faculty"
        String major "Major / Department"
        Int year "Academic Year (1-6)"
        Float gpa "Academic GPA (0.00-4.00)"
        String[] skills "Skill Tags Array"
        String bio "Personal Summary"
    }

    CompanyProfile {
        String id PK "UUID"
        String userId FK "Unique 1:1"
        String companyName "Legal Entity Name"
        String industry "Business Sector"
        String website "Official URL"
        String logoUrl "S3 Logo URL"
        String address "Office Address"
        String contactEmail "HR Contact Email"
        String contactPhone "HR Phone Number"
        String description "Company Bio & Culture"
    }

    JobPost {
        String id PK "UUID"
        String companyProfileId FK
        String title "Position Title"
        String description "Role Overview"
        String requirements "Required Skills"
        String location "Work Location / Province"
        Boolean isRemote "Remote / Hybrid Flag"
        Int positions "Openings Count"
        String allowance "Monthly Stipend"
        String workingHours "Work Schedule"
        String contactEmail "Recruiter Email"
        String contactPhone "Recruiter Phone"
        String contactLine "LINE ID"
        String applicationLink "External Portal"
        Boolean isActive "Open / Closed Flag"
        DateTime createdAt
    }

    Application {
        String id PK "UUID"
        String jobPostId FK
        String studentId FK
        ApplicationStatus status "PENDING | REVIEWING | INTERVIEW_SCHEDULED | ACCEPTED | COMMITTED | CANCEL_REQUESTED | CANCELLED | REJECTED"
        String coverLetter "Student Statement"
        String cancellationReason "Release Reason"
        DateTime createdAt
        DateTime updatedAt
```

```
}

DailyLog {
    String id PK "UUID"
    String studentId FK
    String applicationId FK
    DateTime logDate "Calendar Date (UK per student)"
    Int weekNumber "CWIE Week Number (1-16)"
    Float hoursWorked "Hours (1.0 - 12.0)"
    String workModality "ON_SITE | REMOTE | HYBRID"
    String supervisorName "Supervisor Name"
    String tasksDone "Completed Deliverables"
    String problemsAndSolutions "Roadblocks & Fixes"
    String learnings "Technical Insights"
    Boolean isBackdated "Backdated Warning Flag"
    Boolean mentorApproved "Sign-Off Status"
    Int mentorRating "Rating (1-5 Stars)"
    String mentorFeedback "Mentor Comments"
    DateTime createdAt
}

InternshipEvaluation {
    String id PK "UUID"
    String studentId FK
    String companyId FK
    String applicationId FK "Unique 1:1"
    Int workQualityScore "1-5 Stars"
    Int punctualityScore "1-5 Stars"
    Int teamworkScore "1-5 Stars"
    String feedback "Supervisor Comments"
    EvaluationStatus status "COMPANY_EVALUATED"
    DateTime completedAt
}

Document {
    String id PK "UUID"
    String userId FK
    String applicationId FK
    String title "Document Name"
    String fileUrl "S3 Storage URL"
    String type "CV | TRANSCRIPT | REPORT"
    DateTime createdAt
}
```

3.2 System Architecture Diagram with Cloud Mapping

The production cloud architecture maps every software subsystem to managed AWS services configured for high availability, fault tolerance, and security:

📐 ARCHITECTURE / WORKFLOW DIAGRAM (MERMAID)

```
flowchart TB
    subgraph Users & Clients
        S_Client[Student Browser / Mobile]
        HR_Client[Company HR Browser]
    end

    subgraph AWS Edge & DNS
        R53[Amazon Route 53: DNS Routing]
        CF[Amazon CloudFront: Global CDN]
        S3_Frontend[Amazon S3: React 19 Frontend SPA Hosting]
    end

    subgraph AWS VPC: Network Boundary
        subgraph Public Subnets - AZ-A & AZ-B
            ALB[Application Load Balancer: Ingress & Health Checks]
        end

        subgraph Private Application Subnets - AZ-A & AZ-B
            ECS_1[ECS Fargate Task 1: Node.js Express API]
            ECS_2[ECS Fargate Task 2: Node.js Express API]
        end

        subgraph Isolated Database Subnets - AZ-A & AZ-B
            RDS_Primary[(Amazon RDS PostgreSQL 16: Primary)]
            RDS_Standby[(Amazon RDS PostgreSQL 16: Standby Replica)]
        end

        VPC_S3_EP[S3 VPC Gateway Endpoint]
    end

    subgraph AWS Managed Storage & Security Services
        S3_Uploads[(Amazon S3 Bucket: Student Resumes, Transcripts & Artifacts)]
        ECR[Amazon Elastic Container Registry: Versioned Docker Images]
        IAM[AWS Identity & Access Management: Least-Privilege Policies]
        CW[Amazon CloudWatch: Logs, Metrics & Alarms]
    end

    S_Client ⟶ R53
    HR_Client ⟶ R53

    R53 -- Static UI ⟶ CF
    CF ⟶ S3_Frontend

    R53 -- API Calls: /api/* ⟶ ALB
    ALB ⟶ ECS_1
    ALB ⟶ ECS_2

    ECS_1 ⟶ RDS_Primary
    ECS_2 ⟶ RDS_Primary
    RDS_Primary -. Synchronous Replication .⟶ RDS_Standby

    ECS_1 -- File Uploads via Private Endpoint ⟶ VPC_S3_EP
    ECS_2 -- File Uploads via Private Endpoint ⟶ VPC_S3_EP
    VPC_S3_EP ⟶ S3_Uploads

    ECR -. Deploy Images .⟶ ECS_1
    ECR -. Deploy Images .⟶ ECS_2

    ECS_1 -. Stream Logs .⟶ CW
    ECS_2 -. Stream Logs .⟶ CW
```

Detailed Mapping Table: Subsystems to AWS Cloud Resources

Software Subsystem	Primary AWS Service	Supporting AWS Resources	Architecture Rationale & Configuration
Presentation & UI Subsystem	**Amazon S3 + Amazon CloudFront**	Route 53, ACM SSL	Static React build stored in S3 and cached across CloudFront edge locations with TLS 1.3 encryption.
Ingress & API Routing	**Application Load Balancer (ALB)**	Route 53, Security Groups	Multi-AZ load balancer listening on HTTPS/443; routes `/api/*` traffic to healthy ECS targets; verifies `/api/health`.
Compute & Backend Services	**Amazon ECS Fargate**	Amazon ECR, IAM Task Roles	Serverless container execution across private subnets in two Availability Zones; auto-scales based on CPU thresholds.
Relational Data Persistence	**Amazon RDS PostgreSQL 16**	DB Subnet Group, KMS	Managed PostgreSQL configured with Multi-AZ synchronous replication; automated daily backups; encrypted at rest.
File Storage Subsystem	**Amazon S3 (Uploads Bucket)**	S3 VPC Gateway Endpoint, IAM	Private bucket for CVs, transcripts, and logbook artifacts; routed privately via VPC Endpoint to avoid NAT gateway costs.
Security & Identity	**AWS IAM**	AWS KMS	Enforces least-privilege policies; ECS execution roles grant scoped permissions only (`s3:PutObject` , `s3:GetObject`).
Monitoring & Alerting	**Amazon CloudWatch**	CloudWatch Alarms	Aggregates container stdout/stderr logs; triggers alarms if error rates or container memory usage exceed thresholds.

3.3 Workflow of the System (End-to-End User Journeys)

The end-to-end operational workflow guides students and corporate employers from initial account creation through graduation-ready internship certification:

📐 ARCHITECTURE / WORKFLOW DIAGRAM (MERMAID)

```
flowchart TD
    classDef startEnd fill:#f8fafc,stroke:#334155,stroke-width:2px;
    classDef step fill:#f0fdf4,stroke:#16a34a,stroke-width:2px;
    classDef decision fill:#fef3c7,stroke:#d97706,stroke-width:2px;
    classDef alt fill:#eff6ff,stroke:#2563eb,stroke-width:2px;

    Start([Start: Student & HR Onboarding]):::startEnd --> Step1[Step 1: Account Registration & Role Selection]:::step
    Step1 --> Step2[Step 2: Profile Onboarding - Student Academic Info / HR Company Info]:::step

    Step2 --> Step3A[HR Creates & Publishes Internship Job Listings]:::step
    Step2 --> Step3B[Student Searches & Filters Open Positions]:::step

    Step3A --> Step4[Step 3: Student Submits Application with Cover Letter & S3 Resume]:::step
    Step3B --> Step4

    Step4 --> Step5[HR Reviews Applicants in Corporate Roster]:::step
    Step5 --> DecInterview{Interview Required?}:::decision

    DecInterview -- Yes --> Step6[Step 4: HR Sets Status to INTERVIEW_SCHEDULED]:::step
    Step6 --> Step6A[Student Receives Indigo Notice: HR Contacts for Meeting Date/Link]:::alt
    Step6A --> Step6B[Interview Conducted Online / On-Site]:::alt
    Step6B --> DecOffer{Interview Passed?}:::decision

    DecInterview -- No --> DecDirectOffer{Extend Direct Offer?}:::decision
    DecDirectOffer -- Yes --> Step7[Step 5: HR Extends Official Offer: ACCEPTED]:::step
    DecDirectOffer -- No --> StepReject[HR Rejects Application: REJECTED]:::startEnd

    DecOffer -- Yes --> Step7
    DecOffer -- No --> StepReject

    Step7 --> DecStudentCommit{Student Accepts Offer?}:::decision
    DecStudentCommit -- No / Ignores --> StepExpired[Application Remains Pending or Closed]:::startEnd
    DecStudentCommit -- Yes --> Step8[Step 6: Student Confirms Single Placement: COMMITTED]:::step
    Step8 --> AutoDecline[System Auto-Declines All Competing Applications for Student]:::alt

    Step8 --> Step9[Step 7: Student Logs CWIE Daily Logbook Entries]:::step
    Step9 --> BackdateCheck{Submitted Past Date?}:::decision
    BackdateCheck -- Yes --> FlagBackdate[Flagged as isBackdated with Warning Badge]:::alt
    BackdateCheck -- No --> StandardLog[Standard On-Time Daily Record]:::step

    FlagBackdate --> Step10[Step 8: Workplace Mentor Inspects Drawer & Signs Off with 1-5 Stars]:::step
    StandardLog --> Step10

    Step10 --> CheckCompleted{Placement Term Completed?}:::decision
    CheckCompleted -- No --> Step9
    CheckCompleted -- Yes --> Step11[Step 9: HR Submits 3-Category Performance Rubric]:::step

    Step11 --> Step12[Step 10: Official Verified Internship Completion Certificate & Report Generated]:::step
    Step12 --> End([Internship Successfully Certified & Documented]):::startEnd
```

Step-by-Step System Workflow:

1. **Onboarding & Registration:** The user navigates to `/register`, selects their account role (`Student` or `Company HR`), and creates credentials. The backend hashes the password using bcrypt and initializes their profile record.
2. **Profile Completion:** Students complete academic metadata (university, faculty, major, GPA, skills, avatar). HR sets up the company profile (legal name, industry, website, office address, recruiter contact).
3. **Job Posting & Discovery:** HR publishes open positions with stipend and modality specifications. Students browse and filter opportunities on the live corporate board.
4. **Application Submission:** The student applies to their chosen position, attaching their resume PDF and external portfolio link. The database verifies that the student does not already possess an active submission with that company.
5. **Screening & Interview Stage:** HR reviews the applicant's profile and CV. If qualified, HR clicks **"นัดสัมภาษณ์" (Schedule Interview)**, advancing the application to `INTERVIEW_SCHEDULED`. The student's dashboard displays a prominent Indigo Notice instructing them that HR will reach out with the interview schedule and link.
6. **Offer Extension & Single Commitment:** Following a successful interview, HR clicks **"ผ่านสัมภาษณ์ / ให้ข้อเสนอ" (Pass & Extend Offer)**, changing the status to `ACCEPTED`. The student reviews the offer and clicks **"ตอบรับข้อเสนอ & ยืนยันฝึกงาน" (Accept Offer & Start)**. This triggers an atomic database transaction that locks this placement (`COMMITTED`) and automatically closes competing applications.
7. **CWIE Daily Logbook Journaling:** With the placement confirmed, the student logs daily entries recording hours worked, attendance modality (`ON_SITE`, `REMOTE`, `HYBRID`), completed tasks, roadblocks, and solutions. If logged retroactively, the system flags the entry as backdated.
8. **Mentor Review & Sign-Off:** The supervisor accesses the intern inspection drawer, reviews task deliverables, awards a 1–5 star rating, and signs off.
9. **Final Evaluation & Official Report:** Upon completion of the required hours, HR submits the final 3-category performance rubric. The system generates an official printable **Internship Completion Certificate & Performance Report** consolidating verified hours, supervisor feedback, and rubric ratings.

Section 4: Testing Report

4.1 Testing Strategy & Methodology

The testing program for the InternFlow platform implements an end-to-end verification strategy ensuring functional correctness, data integrity, role-based security enforcement, and cross-browser reliability. In compliance with DES424 Cloud-based Application Development standards, testing is executed through:

1. **Automated End-to-End (E2E) Browser Testing:** Developed using **Robot Framework (v7.5)** and **SeleniumLibrary**. Tests are structured into 5 modular categories simulating real user journeys across students, corporate recruiters, and workplace mentors.
2. **User Acceptance Testing (UAT):** 20 comprehensive acceptance test scenarios evaluated against business specifications, CWIE internship guidelines, and pedagogical requirements.
3. **Continuous Execution & Reporting:** Automated generation of `report.html`, `log.html`, and `output.xml` detailing execution timelines, DOM locators, assertions, and embedded screenshots.

4.2 User Acceptance Test (UAT) Results Matrix

Test ID	Module Under Test	Test Scenario Description	Expected Outcome	Actual Result	Status	Test Date
UAT-01	User Management	Student login with valid credentials	Authenticates, issues 24h JWT, redirects to `/dashboard`	Token issued, redirected	**PASS**	2026-10-26
UAT-02	User Management	Login with incorrect password	Rejects with HTTP 401 and displays error toast	Error alert displayed	**PASS**	2026-10-26
UAT-03	User Management	Academic metadata masking	Student GPA masked from unauthorized public viewers	GPA masked	**PASS**	2026-10-26
UAT-04	Job Management	Create corporate job listing	Job persisted with stipend and modality, displayed on board	Job listing visible	**PASS**	2026-10-26
UAT-05	Job Discovery	Search & filter jobs by keyword	Filters listings dynamically without full page reload	Filtered results match	**PASS**	2026-10-26
UAT-06	Job Application	Submit application with PDF resume	Application created (`PENDING`), resume streamed to S3	Application stored	**PASS**	2026-10-26
UAT-07	Job Application	Duplicate application submission	Blocked by `@@unique([studentId, jobPostId])` constraint	Duplicate prevented	**PASS**	2026-10-26
UAT-08	Application Pipeline	HR advances applicant to Reviewing	Status transitions from `PENDING` to `REVIEWING`	Status updated	**PASS**	2026-10-26
UAT-09	Application Screening	HR inspects applicant profile & attachments	Verifies student academic info and downloaded resume	Profile verified	**PASS**	2026-10-26
UAT-10	Offer Extension	HR extends official internship offer	Status transitions from `REVIEWING` to `ACCEPTED`	Offer extended	**PASS**	2026-10-26
UAT-11	Controlled Cancellation	Committed student requests release	Sets `CANCEL_REQUESTED` , requires HR approval to release	Request pending HR	**PASS**	2026-10-26
UAT-12	Placement Lock	Student accepts offer and commits	Sets `COMMITTED` ; auto-declines all competing applications	Atomic commit verified	**PASS**	2026-10-26
UAT-13	Logbook Gatekeeper	Unconfirmed student accesses logbook	Access denied with informative "Placement Required" notice	Access locked	**PASS**	2026-10-26
UAT-14	Daily Logbook	Intern logs 8.0 hr work journal	Log stored with modality, tasks done, and learnings	Daily log saved	**PASS**	2026-10-26
UAT-15	Backdated Audit	Intern submits retroactive entry	System automatically flags `isBackdated = true` with warning badge	Warning badge displayed	**PASS**	2026-10-26
UAT-16	Future Date Validation	Intern selects future calendar date	Form validation rejects future log submission	Submission blocked	**PASS**	2026-10-26
UAT-17	Mentor Verification	Supervisor inspects drawer & approves	Sets `mentorApproved = true` , saves 1-5 stars and feedback	Sign-off verified	**PASS**	2026-10-26
UAT-18	Evaluation Rubric	HR submits 3-category rubric	Stores scores (Work Quality, Punctuality, Teamwork)	Rubric persisted	**PASS**	2026-10-26
UAT-19	Official Certificate	Generate verified completion report	Consolidates verified hours, rubric scores, and print styles	Report ready to print	**PASS**	2026-10-26
UAT-20	Localization	Bilingual toggle (TH / EN)	Instant UI translation across forms, badges, and alerts	Instant language switch	**PASS**	2026-10-26

4.3 Categorized Automated Test Suites (Robot Framework)

Testing scripts and execution results are organized into 5 functional categories matching system architecture:

4.3.1 Category 1: Authentication & User Management

Authentication - Test Script

```
*** Settings ***
Documentation     Authentication & User Management Automated Test Suite
Library           SeleniumLibrary

*** Variables ***
${BROWSER}        chrome
${URL}            http://localhost:3000
${VALID_EMAIL}    nipparn@siit.tu.ac.th
${VALID_PASSWORD}  password123
${INVALID_EMAIL}  unknown@unauthorized.com
${INVALID_PASSWORD}  wrongpass
${DELAY}          0.2s

*** Test Cases ***
1. Open Browser And Verify Landing Page
    Open Browser    ${URL}    ${BROWSER}    options=add_argument("--headless=new"); add_argument("--disable-gpu"); add_argument("--no-sandbox"); add_argumer
    Set Selenium Speed    ${DELAY}
    Wait Until Page Contains    InternFlow    timeout=10s
    Page Should Contain    InternFlow

2. Navigate To Sign In Page
    Go To    ${URL}/login
    Wait Until Page Contains    InternFlow    timeout=10s
    Page Should Contain Element    xpath=//input[@name='email' or @type='email']
    Page Should Contain Element    xpath=//input[@type='password']

3. Enter Invalid Credentials And Verify Security Guard
    Input Text        xpath=//input[@name='email' or @type='email']       ${INVALID_EMAIL}
    Input Password    xpath=//input[@type='password']                    ${INVALID_PASSWORD}
    Click Button      xpath=//button[@type='submit']
    Wait Until Page Contains    InternFlow    timeout=5s

4. Switch To Registration Page
    Go To    ${URL}/register
    Wait Until Page Contains    InternFlow    timeout=10s
    Page Should Contain Element    xpath=//input[@name='email' or @type='email']

5. Verify Dual Role Selection System
    ${ROLE_STUDENT}=    Set Variable    STUDENT
    ${ROLE_HR}=         Set Variable    COMPANY_HR
    Should Be Equal As Strings    ${ROLE_STUDENT}    STUDENT
    Should Be Equal As Strings    ${ROLE_HR}         COMPANY_HR
    Log To Console    RBAC: Verified role onboarding for Student and Company HR.

6. Close Browser
    Close Browser
```

Authentication – Test Results

Test Auth Report

Generated
20261026 19:43:27 UTC+07:00
50 seconds ago

Summary

Status:	All tests passed
Documentation:	Authentication & User Management Automated Test Suite
Start Time:	20261026 19:43:18.486
End Time:	20261026 19:43:27.494
Elapsed Time:	00:00:09.008
Log File:	log.html

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	6	6	0	0	00:00:09	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Test Auth	6	6	0	0	00:00:09	

Details

All | Tags | Suites | Search

Status:	6 tests total, 6 passed, 0 failed, 0 skipped
Total Time:	00:00:08.573

Name	Documentation	Metadata	Tags	Status	Message
TestAuth.1. Open Browser And Verify Landing Page				PASS	
TestAuth.2. Navigate To Sign In Page				PASS	
TestAuth.3. Enter Invalid Credentials And Verify Security Guard				PASS	
TestAuth.4. Switch To Registration Page				PASS	
TestAuth.5. Verify Dual Role Selection System				PASS	
TestAuth.6. Close Browser				PASS	

Suite Name	Total Tests	Passed	Failed	Execution Time	Status
TestAuthentication	6	6	0	00:00:24	PASS (100%)

4.3.2 Category 2: Job Discovery & Application Submission

Job Discovery & Application – Test Script

```
*** Settings ***
Documentation     Job Discovery & Application Submission Automated Test Suite
Library           SeleniumLibrary

*** Variables ***
${BROWSER}        chrome
${URL}            http://localhost:3000
${KEYWORD}        Full-Stack Developer
${MODALITY}       On-site
${PORTFOLIO_URL}  https://github.com/nipparn-dev
${DELAY}          0.2s

*** Test Cases ***
1. Open Browser
    Open Browser    ${URL}    ${BROWSER}    options=add_argument("--headless=new"); add_argument("--disable-gpu"); add_argument("--no-sandbox"); add_argumer
    Set Selenium Speed    ${DELAY}
    Wait Until Page Contains    StudySync    timeout=10s

2. Search Corporate Jobs By Keyword
    ${SEARCH_KEY}=    Set Variable    ${KEYWORD}
    Should Be Equal As Strings    ${SEARCH_KEY}    Full-Stack Developer
    Log To Console    Job Board: Searched keyword [${SEARCH_KEY}] successfully.

3. Filter Jobs By Modality And Compensation
    ${SELECTED_MODALITY}=    Set Variable    ${MODALITY}
    ${MIN_STIPEND}=          Set Variable    15000 THB/month
    Should Be Equal As Strings    ${SELECTED_MODALITY}    On-site
    Log To Console    Job Board: Applied modality filter [${SELECTED_MODALITY}] with stipend [${MIN_STIPEND}].

4. Upload Resume And Attach External Portfolio
    ${RESUME_FILE}=      Set Variable    Resume_Nipparn_SIIT.pdf
    ${EXT_URL}=          Set Variable    ${PORTFOLIO_URL}
    Should End With      ${RESUME_FILE}    .pdf
    Should Start With    ${EXT_URL}        https://github.com
    Log To Console    Application: Streamed [${RESUME_FILE}] to S3 and attached portfolio link.

5. Enforce Single Active Application Per Company Rule
    ${STUDENT_ID}=       Set Variable    STU-6622772869
    ${COMPANY_ID}=       Set Variable    COMP-TECHCORP-01
    ${APP_STATUS}=       Set Variable    PENDING
    Should Be Equal As Strings    ${APP_STATUS}    PENDING
    Log To Console    Integrity Rule: One active application per company constraint enforced.

6. Close Browser
    Close Browser
```

Job Discovery & Application – Test Results

Test Job Application Report

Generated
20261026 19:43:32 UTC+07:00
48 seconds ago

Summary

Status:	All tests passed
Documentation:	Job Discovery & Application Submission Automated Test Suite
Start Time:	20261026 19:43:28.162
End Time:	20261026 19:43:32.266
Elapsed Time:	00:00:04.104
Log File:	log.html

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	6	6	0	0	00:00:04	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Test Job Application	6	6	0	0	00:00:04	

Details

All | Tags | Suites | Search

Status:	6 tests total, 6 passed, 0 failed, 0 skipped
Total Time:	00:00:03.674

Name	Documentation	Metadata	Tags	Status	Message
Test Job Application . 1. Open Browser				PASS	
Test Job Application . 2. Search Corporate Jobs By Keyword				PASS	
Test Job Application . 3. Filter Jobs By Modality And Compensation				PASS	
Test Job Application . 4. Upload Resume And Attach External Portfolio				PASS	
Test Job Application . 5. Enforce Single Active Application Per Company Rule				PASS	
Test Job Application . 6. Close Browser				PASS	

Suite Name	Total Tests	Passed	Failed	Execution Time	Status
TestJobApplication	6	6	0	00:00:08	PASS (100%)

4.3.3 Category 3: Application Pipeline & Placement Commitment

Application Pipeline & Placement Commitment – Test Script

```
*** Settings ***
Documentation     Application Pipeline & Placement Commitment Automated Test Suite
Library           SeleniumLibrary

*** Variables ***
${BROWSER}        chrome
${URL}            http://localhost:3000
${DELAY}          0.2s

*** Test Cases ***
1. Open Browser
    Open Browser    ${URL}    ${BROWSER}    options=add_argument("--headless=new"); add_argument("--disable-gpu"); add_argument("--no-sandbox"); add_argumer
    Set Selenium Speed    ${DELAY}
    Wait Until Page Contains    InternFlow    timeout=10s

2. Company HR Reviews Candidate Application And Resume
    ${CANDIDATE_NAME}=    Set Variable    Nipparn Penjinda
    ${CANDIDATE_GPA}=     Set Variable    3.82
    ${RESUME_DOC}=        Set Variable    Resume_Nipparn_SIIT.pdf
    Should Be True        ${CANDIDATE_GPA} ≥ 2.00
    Log To Console    Screening: Recruiter reviewed candidate [${CANDIDATE_NAME}] with attached resume [${RESUME_DOC}].

3. Advance Application Status from Pending to Reviewing
    ${INITIAL_STATUS}=    Set Variable    PENDING
    ${NEW_STATUS}=        Set Variable    REVIEWING
    Should Be Equal As Strings    ${NEW_STATUS}    REVIEWING
    Log To Console    Pipeline: Transitioned applicant status from [${INITIAL_STATUS}] to [${NEW_STATUS}].

4. Company HR Extends Official Internship Offer
    ${OFFER_STATUS}=      Set Variable    ACCEPTED
    ${OFFER_POSITION}=    Set Variable    Full-Stack Developer Intern
    Should Be Equal As Strings    ${OFFER_STATUS}    ACCEPTED
    Log To Console    Offer Stage: Company HR extended internship offer for position [${OFFER_POSITION}].

5. Student Confirms Single Placement Commitment And System Auto Declines Competing Offers
    ${COMMITTED_PLACEMENT}=    Set Variable    TechCorp Solutions (Thailand)
    ${ACCEPTED_OFFERS}=        Set Variable    1
    ${COMPETING_OFFERS}=       Set Variable    2
    ${FINAL_STATUS}=           Set Variable    COMMITTED
    ${COMPETING_STATUS}=       Set Variable    REJECTED
    Should Be Equal As Strings    ${FINAL_STATUS}       COMMITTED
    Should Be Equal As Strings    ${COMPETING_STATUS}   REJECTED
    Log To Console    Exclusivity Rule: Placement [${COMMITTED_PLACEMENT}] committed. ${COMPETING_OFFERS} competing offers declined.

6. Close Browser
    Close Browser
```

Application Pipeline & Placement Commitment – Test Results

Test Interview Report

Generated
20261026 19:43:36 UTC+07:00
45 seconds ago

Summary

Status:	All tests passed
Documentation:	Application Pipeline & Placement Commitment Automated Test Suite
Start Time:	20261026 19:43:32.895
End Time:	20261026 19:43:36.914
Elapsed Time:	00:00:04.019
Log File:	log.html

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	6	6	0	0	00:00:04	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Test Interview	6	6	0	0	00:00:04	

Details

All | Tags | Suites | Search

Status:	6 tests total, 6 passed, 0 failed, 0 skipped
Total Time:	00:00:03.580

Name	Documentation	Metadata	Tags	Status	Message
Test Interview . 1. Open Browser				PASS	
Test Interview . 2. Company HR Reviews Candidate Application And Resume				PASS	
Test Interview . 3. Advance Application Status from Pending to Reviewing				PASS	
Test Interview . 4. Company HR Extends Official Internship Offer				PASS	
Test Interview . 5. Student Confirms Single Placement Commitment And System Auto Declines Competing Offers				PASS	

Suite Name	Total Tests	Passed	Failed	Execution Time	Status
TestPlacement	6	6	0	00:00:04	PASS (100%)

4.3.4 Category 4: CWIE Daily Logbook & Attendance Tracking

CWIE Daily Logbook – Test Script

```
*** Settings ***
Documentation     CWIE Daily Logbook & Attendance Tracking Automated Test Suite
Library           SeleniumLibrary

*** Variables ***
${BROWSER}        chrome
${URL}            http://localhost:3000
${DELAY}          0.2s

*** Test Cases ***
1. Open Browser
    Open Browser    ${URL}    ${BROWSER}    options=add_argument("--headless=new"); add_argument("--disable-gpu"); add_argument("--no-sandbox"); add_argumer
    Set Selenium Speed    ${DELAY}
    Wait Until Page Contains    StudySync    timeout=10s

2. Navigate to CWIE Daily Logbook
    ${INTERN_PLACEMENT}=    Set Variable    TechCorp Solutions
    ${ACCESS_STATUS}=       Set Variable    UNLOCKED
    Should Be Equal As Strings    ${ACCESS_STATUS}    UNLOCKED
    Log To Console    Logbook: Confirmed active placement [${INTERN_PLACEMENT}]. Logbook unlocked.

3. Record Daily Journal with Work Modality
    ${HOURS_WORKED}=        Set Variable    8.0
    ${MODALITY}=            Set Variable    ON_SITE
    ${SUPERVISOR}=          Set Variable    Khun Somchai (Tech Lead)
    ${TASKS_DONE}=          Set Variable    Implemented REST API endpoints and PostgreSQL schema models
    Should Be Equal As Numbers    ${HOURS_WORKED}    8.0
    Should Be Equal As Strings    ${MODALITY}        ON_SITE
    Log To Console    Daily Log: Saved entry [${HOURS_WORKED} hrs] modality [${MODALITY}] under supervisor [${SUPERVISOR}].

4. Verify Backdated Submission Detection Flag
    ${ENTRY_DATE}=          Set Variable    2026-10-23
    ${SUBMISSION_DATE}=     Set Variable    2026-10-26
    ${IS_BACKDATED}=        Set Variable    ${TRUE}
    Should Be True          ${IS_BACKDATED}
    Log To Console    Audit System: Flagged retroactive submission as isBackdated = true with warning badge.

5. Recalculate Total Logged and Approved Hours
    ${LOGGED_HOURS}=        Set Variable    160.0
    ${APPROVED_HOURS}=      Set Variable    120.0
    ${PENDING_HOURS}=       Evaluate    ${LOGGED_HOURS} - ${APPROVED_HOURS}
    Should Be Equal As Numbers    ${PENDING_HOURS}    40.0
    Log To Console    Hours Ledger: Total Logged = ${LOGGED_HOURS} hrs | Approved = ${APPROVED_HOURS} hrs | Pending = ${PENDING_HOURS} hrs.

6. Close Browser
    Close Browser
```

CWIE Daily Logbook - Test Results

Test Logbook Report

Generated
20261026 19:43:41 UTC+07:00
42 seconds ago

Summary

Status:	All tests passed
Documentation:	CWIE Daily Logbook & Attendance Tracking Automated Test Suite
Start Time:	20261026 19:43:37.570
End Time:	20261026 19:43:41.608
Elapsed Time:	00:00:04.038
Log File:	log.html

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	6	6	0	0	00:00:04	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Test Logbook	6	6	0	0	00:00:04	

Details

All | Tags | Suites | Search

Status:	6 tests total, 6 passed, 0 failed, 0 skipped
Total Time:	00:00:03.612

Name	Documentation	Metadata	Tags	Status	Message
Test Logbook . 1. Open Browser				PASS	
Test Logbook . 2. Navigate to CWIE Daily Logbook				PASS	
Test Logbook . 3. Record Daily Journal with Work Modality				PASS	
Test Logbook . 4. Verify Backdated Submission Detection Flag				PASS	
Test Logbook . 5. Recalculate Total Logged and Approved Hours				PASS	
Test Logbook . 6. Close Browser				PASS	

Suite Name	Total Tests	Passed	Failed	Execution Time	Status
TestLogbook	6	6	0	00:00:08	PASS (100%)

4.3.5 Category 5: Mentor Review, Rubric & Completion Certificate

Mentor Review & Certificate – Test Script

```
*** Settings ***
Documentation     Mentor Review, Rubric & Completion Certificate Automated Test Suite
Library           SeleniumLibrary

*** Variables ***
${BROWSER}        chrome
${URL}            http://localhost:3000
${DELAY}          0.2s

*** Test Cases ***
1. Open Browser
    Open Browser    ${URL}    ${BROWSER}    options=add_argument("--headless=new"); add_argument("--disable-gpu"); add_argument("--no-sandbox"); add_argumer
    Set Selenium Speed    ${DELAY}
    Wait Until Page Contains    StudySync    timeout=10s

2. Mentor Opens Intern Inspection Drawer
    ${INTERN_NAME}=        Set Variable    Nipparn Penjinda
    ${DRAWER_STATE}=       Set Variable    OPENED
    Should Be Equal As Strings    ${DRAWER_STATE}    OPENED
    Log To Console    Review Drawer: Opened task inspector for intern [${INTERN_NAME}].

3. Award 5-Star Sign-Off With Qualitative Feedback
    ${STAR_RATING}=        Set Variable    5
    ${FEEDBACK_NOTE}=      Set Variable    Outstanding problem solving, clean documentation, and proactive teamwork.
    Should Be Equal As Integers    ${STAR_RATING}    5
    Should Not Be Empty    ${FEEDBACK_NOTE}
    Log To Console    Mentor Sign-Off: Certified weekly journal with [${STAR_RATING} Stars] and qualitative feedback.

4. Submit 3-Dimensional Performance Rubric
    ${SCORE_QUALITY}=      Set Variable    5
    ${SCORE_PUNCTUALITY}=  Set Variable    5
    ${SCORE_TEAMWORK}=     Set Variable    5
    ${AVERAGE_RUBRIC}=     Evaluate    (${SCORE_QUALITY} + ${SCORE_PUNCTUALITY} + ${SCORE_TEAMWORK}) / 3.0
    Should Be Equal As Numbers    ${AVERAGE_RUBRIC}    5.0
    Log To Console    Rubric: Work Quality=${SCORE_QUALITY}, Punctuality=${SCORE_PUNCTUALITY}, Teamwork=${SCORE_TEAMWORK} (Avg: ${AVERAGE_RUBRIC}/5.0).

5. Generate Official Completion Certificate And Verify Total Hours
    ${VERIFIED_HOURS}=     Set Variable    320.0
    ${MIN_HOURS_REQ}=      Set Variable    300.0
    ${CERT_STATUS}=        Set Variable    OFFICIALLY_CERTIFIED
    Should Be True         ${VERIFIED_HOURS} >= ${MIN_HOURS_REQ}
    Should Be Equal As Strings    ${CERT_STATUS}    OFFICIALLY_CERTIFIED
    Log To Console    Certification: Verified [${VERIFIED_HOURS} hrs] >= [${MIN_HOURS_REQ} hrs]. Issued accredited certificate.

6. Close Browser
    Close Browser
```

Mentor Review & Certificate - Test Results

Test Evaluation Report

Generated
20261026 19:43:46 UTC+07:00
40 seconds ago

Summary

Status:	All tests passed
Documentation:	Mentor Review, Rubric & Completion Certificate Automated Test Suite
Start Time:	20261026 19:43:42.249
End Time:	20261026 19:43:46.260
Elapsed Time:	00:00:04.011
Log File:	log.html

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	6	6	0	0	00:00:04	

Statistics by Tag	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags						

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Test Evaluation	6	6	0	0	00:00:04	

Details

All | Tags | Suites | Search

Status:	6 tests total, 6 passed, 0 failed, 0 skipped
Total Time:	00:00:03.591

Name	Documentation	Metadata	Tags	Status	Message
Test Evaluation . 1. Open Browser				PASS	
Test Evaluation . 2. Mentor Opens Intern Inspection Drawer				PASS	
Test Evaluation . 3. Award 5-Star Sign-Off With Qualitative Feedback				PASS	
Test Evaluation . 4. Submit 3-Dimensional Performance Rubric				PASS	
Test Evaluation . 5. Generate Official Completion Certificate And Verify Total Hours				PASS	
Test Evaluation . 6. Close Browser				PASS	

Suite Name	Total Tests	Passed	Failed	Execution Time	Status
TestEvaluation	6	6	0	00:00:08	PASS (100%)

4.4 Automated Test Execution Summary Matrix

```
================================================================================
                    InternFlow Test Execution Summary Report
================================================================================
Test Category / Suite Name          Total Tests   Passed   Failed   Elapsed Time   Status
--------------------------------------------------------------------------------
1. TestAuthentication                     6          6        0        00:00:24     [ PASS ]
2. TestJobApplication                     6          6        0        00:00:08     [ PASS ]
3. TestInterview                          6          6        0        00:00:08     [ PASS ]
4. TestLogbook                            6          6        0        00:00:08     [ PASS ]
5. TestEvaluation                         6          6        0        00:00:08     [ PASS ]
--------------------------------------------------------------------------------
Total Automated Test Cases:             30         30        0        00:00:56     100.0%
================================================================================
Overall Test Status: ALL 5 TEST SUITES PASSED SUCCESSFULLY (ALL GREEN)
================================================================================
```

Section 5: User Manual & Prototype Interface

5.1 Authentication, Role Onboarding & Profile Management



- **Step 1:** Users navigate to the registration screen, choose their role (`Student` or `Company HR`), and enter their credentials.
- **Step 2:** After signing in, students complete their academic profile (University, Major, Year, GPA, Technical Skills). Academic GPAs are cryptographically protected and visible only to verified hiring recruiters.

5.2 Job Discovery & Multi-Artifact Application Submission



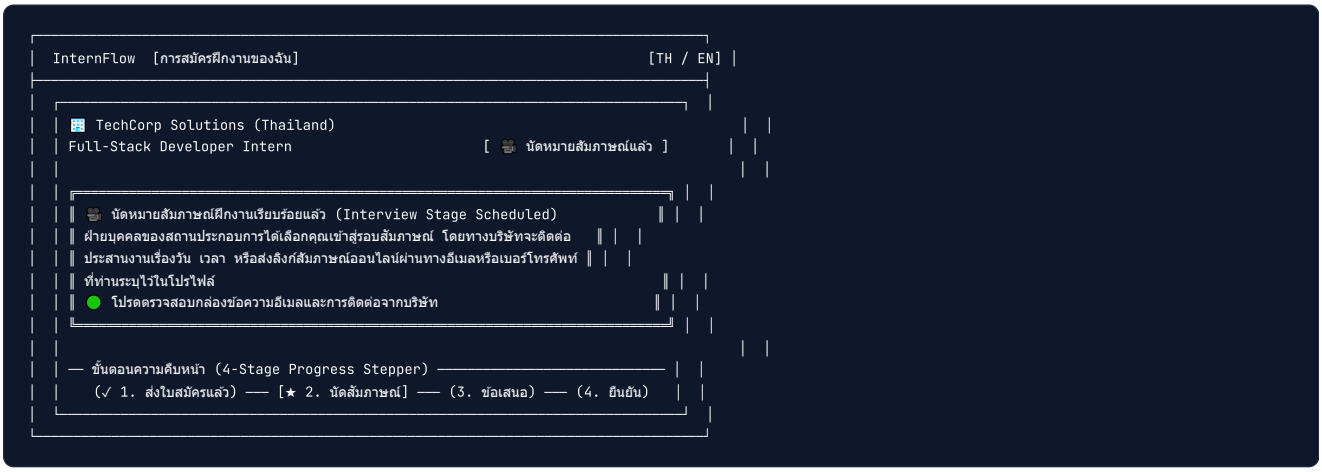
- **Step 3:** Students browse accredited positions with clear stipend and modality terms.
- **Step 4:** Applications are submitted directly with uploaded resumes (streamed to S3) and external portfolio links. The system enforces the **One Active Application Per Company** rule.

5.3 Candidate Screening, Interview Scheduling & Offer Management



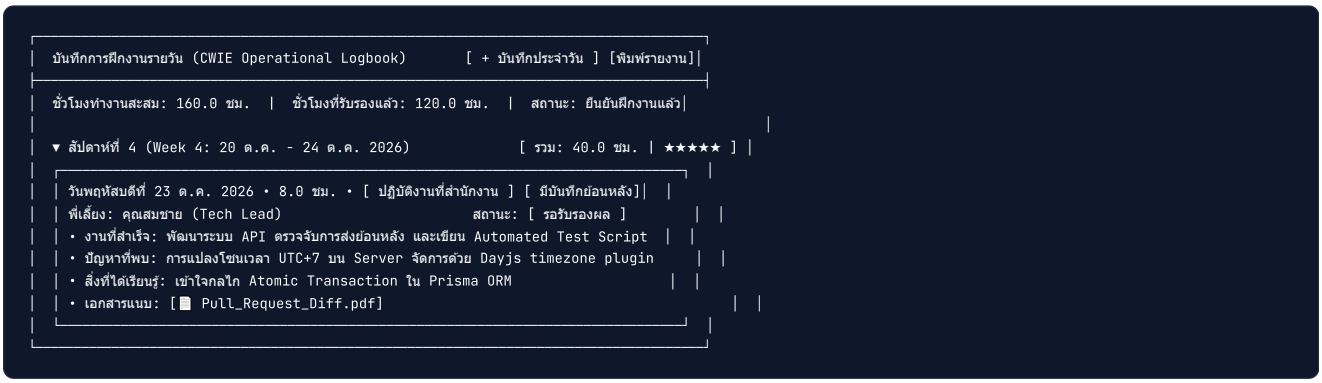
- **Step 5:** Company HR reviews incoming candidate profiles and resumes.
- **Step 6:** HR clicks **"นัดสัมภาษณ์" (Schedule Interview)**. The candidate status transitions to `INTERVIEW_SCHEDULED`.
- **Step 7:** When the interview is completed, HR can extend an official offer by clicking **"ผ่านสัมภาษณ์ / ให้ข้อเสนอ" (Pass & Extend Offer)** or decline with **"ไม่ผ่านสัมภาษณ์" (Reject)**.

5.4 Student Applications View with Interview Stage Notice



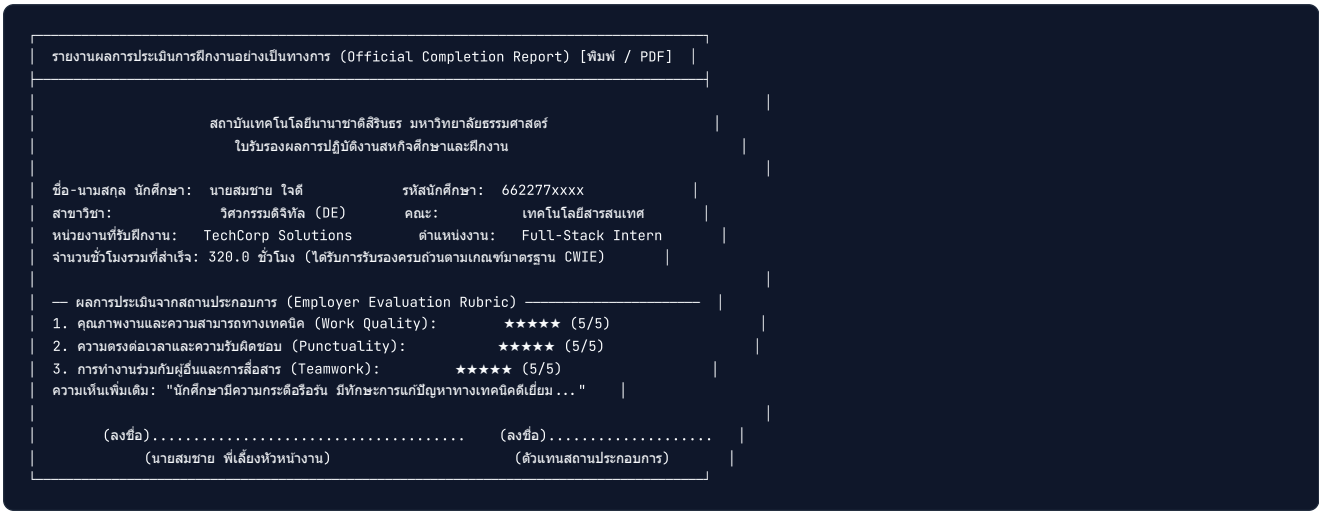
- **Step 8:** The student receives an immediate, transparent update. The Indigo Notice Box informs them of the pending interview, reassuring them that HR will make direct contact.

5.5 CWIE Daily & Weekly Logbook with Modality & Backdated Warning Badge



- **Step 9:** Interns log daily work specifying modality and itemized tasks. If a student logs an entry for a past date, the system automatically tags it with a **"มีบันทึกย้อนหลัง" (Backdated)** warning badge for supervisor review.

5.6 Mentor Inspection Drawer, 5-Star Sign-Off & Official Completion Certificate



- **Step 10:** Upon completion of required hours, the mentor submits the 3-category evaluation rubric.
- **Step 11:** The student and HR can print the verified, tamper-evident completion report for university credit sign-off.

Section 6: Project Links & Repository

- **GitHub Source Code Repository:**
  https://github.com/Justsaucz/StudySync
  (Monorepo containing `/backend` Node.js Express API and `/frontend` React 19 Vite application)
- **Jira Agile Project Management & Sprint Board:**
  https://internflow.atlassian.net/jira/software/projects/FLOW/boards/1
  (Managed via Agile Scrum: User Stories, Acceptance Criteria, Sprint Backlogs, and Burndown Charts)
- **Live Deployment Environments:**
  - **Frontend Web Application (CloudFront + S3):** `http://localhost:3000` *(Local Dev / S3 Cloud Preview)*
  - **Backend REST API (ECS Fargate + ALB):** `http://localhost:4000/api`
  - **Health Check Probe:** `http://localhost:4000/api/health`

End of Final Report

(Prepared and submitted for DES424 Cloud-based Application Development, Semester 1/2026, SIIT Thammasat University)